

Mohamed Mahmoud

Data Engineer | Aspiring MLOps Engineer

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Summary

Computer science student and data engineer experienced in building ETL pipelines, real-time streaming systems, and cloud data architectures — now extending into **MLOps**: training, versioning, and deploying models end-to-end. Recently shipped **AirGuard** (an XGBoost air-quality forecasting service, deployed live) and **Kidney-RAG** (a clinical RAG system deployed on Azure), and co-founded **Sunet**, an ESP32 child-safety wearable.

Technical Skills

- **Languages:** Python, SQL, Java, JavaScript, C#, Node.js
- **Data Engineering:** Apache Spark, Apache Kafka, Apache Airflow, Spark Structured Streaming, ETL/ELT Design, Data Modeling, Data Warehousing
- **Machine Learning / MLOps:** XGBoost, scikit-learn, MLflow, LazyPredict, Feature Engineering, Model Deployment, Time-Series Forecasting, Retrieval-Augmented Generation (RAG)
- **Databases:** PostgreSQL, MongoDB, Cassandra, Elasticsearch, Redis, ChromaDB
- **Cloud & Infrastructure:** AWS (S3, EC2), Azure Container Apps, Docker, Docker Compose, Linux
- **Tools:** Git, GitHub, REST APIs, FastAPI, BeautifulSoup, Kibana, Streamlit
- **Concepts:** Distributed Systems, Stream Processing, OOP, Data Structures, Schema Design

Experience

Sunet | wearsunet.web.app

Co-Founder & Technical Lead

2026 – Present

Alexandria, Egypt

- Leading development of a child-safety wearable on ESP32 (GPS tracking, MPU6050 fall/impact detection, parent-facing companion app) across three completed hardware phases
- Designed the telemetry architecture (FastAPI → Kafka → Spark → Cassandra + Elasticsearch) for real-time location streams and historical event search

National Telecommunication Institute (NTI) × Huawei — AI Track

AI & MLOps Trainee

Aug 2026 – Sep 2026

Egypt (Hybrid)

- Completed an intensive AI track spanning machine learning, data engineering, and MLOps, with hands-on labs across the full model lifecycle (data prep, training, evaluation, deployment, monitoring)
- Built **AirGuard**, the program capstone — an end-to-end MLOps pipeline forecasting Cairo air pollution with an automated health advisory, deployed live (see Projects)

Commercial International Bank (CIB)

Summer Intern

Jul 2026 – Aug 2026

Cairo, Egypt (Hybrid)

- Structured internship covering core banking operations and the role of data across business units; trained in data analysis, data engineering, ML, AI, and cybersecurity; completed the Generative AI module (certificate)

Digital Egypt Pioneers Initiative (DEPI)

Data Engineer Intern

Dec 2025 – Aug 2026

Egypt (Hybrid)

- Built ETL pipelines with Apache Spark and designed data models optimized for analytical workloads on AWS
- Led a six-member team through **NileFlow**, the program capstone, receiving instructor recognition for delivery (see Projects)

Teleperformance

Customer Service Representative

Jul 2025 – Oct 2025

Alexandria, Egypt

- Resolved customer issues across CRM platforms against daily volume/quality targets; surfaced recurring issue patterns to improve resolution scripts

Target for Marketing and Collection

Data Entry Supervisor

2023 – 2024

Alexandria, Egypt

- Supervised a data-entry team on large client datasets; implemented validation checks that reduced downstream data-correction work

Self-Employed — Freelance Discord Bot Developer

Remote

2022 – 2023

- Designed and shipped 100+ custom Discord bots; built RESTful APIs in Node.js/Express with MongoDB and integrated payment and AI APIs for paying clients

Projects

AirGuard — Cairo PM2.5 Forecasting & Health Advisory

NTI × Huawei Capstone

Python, XGBoost, MLflow, Airflow, PostgreSQL, Streamlit, Docker

[GitHub](#) | [Live](#)

- Built an end-to-end MLOps pipeline forecasting Cairo PM2.5 6–12 hours ahead and mapping each forecast to an EPA health advisory; deployed live on Streamlit Community Cloud
- Trained an XGBoost regressor (selected via LazyPredict benchmarking of ~40 models) that beat the persistence baseline by ~35% MAE on a time-ordered holdout
- Versioned models in MLflow with champion-vs-challenger promotion; hourly inference and weekly retraining orchestrated with Airflow; containerized with Docker

Kidney-RAG — Clinical Retrieval-Augmented Generation

Creativa × Orange AI Hackathon, Aug 2026

Python, FastAPI, ChromaDB, MedEmbed, RAG, Docker, Azure

[GitHub](#) | [Live](#)

- Led a 4-person team building a clinical RAG system answering CKD questions strictly from official guidelines (KDIGO, NICE, USPSTF), with inline citations and explicit refusal on out-of-corpus questions
- Hybrid retriever (semantic + BM25, weighted RRF) over 666 chunks with MedEmbed-large in ChromaDB (Hit@5 0.867, faithfulness 1.0); served via FastAPI and deployed to Azure Container Apps

NileFlow — Real-Time Traffic Congestion Intelligence

DEPI Capstone

Kafka, Spark, Airflow, Cassandra, Elasticsearch, Redis, Docker

[GitHub](#)

- Led a 6-member team building a real-time congestion platform for Greater Cairo and Alexandria, ingesting live traffic (TomTom), weather (Open-Meteo), and transit events through three Kafka topics
- Computed a rolling congestion index with Spark Structured Streaming time-window joins; Cassandra (time-series) + Elasticsearch (alerts), Airflow DAGs, and Redis pub/sub to a live Streamlit map

Real-Time Crypto Trade Pipeline

Personal Project

Kafka, Spark, Cassandra, Elasticsearch, Kibana

[GitHub](#)

- Streamed live Binance trades through Kafka, processed with Spark Structured Streaming (tumbling windows + watermarks) into Cassandra and Elasticsearch; unified sinks via foreachBatch and tuned shuffle partitions for lower latency

Egypt Jobs Pipeline

Personal Project (V3 Complete)

Python, Airflow, PostgreSQL, Redis, Docker

[GitHub](#)

- Built a three-version scraping pipeline for Egyptian tech jobs: an Airflow DAG (scheduling, backoff retries, dedup, Streamlit dashboard) and a crash-proof real-time polling layer with Redis SET dedup and role/seniority filters

Certifications

- **Associate Data Engineer** — DataCamp, 2026 (ID 854732) — [Certificate](#)
- **IBM Data Engineering Professional Certificate** — IBM / Coursera, in progress, 2026 — completed [Python for Data Science, AI & Development](#)
- **NTI × Huawei AI Track (AI & MLOps)** — 2026 — [Certificate](#)
- **Generative AI — Enterprise Applications** — CIB Summer Internship, 2026 — [Certificate](#)
- **Database Fundamentals** — Information Technology Institute (ITI), 2025 — [Certificate](#)
- **Python Programming for AI** — Alexandria University, 2024 — [Certificate](#)

Education

Alexandria University — Faculty of Science

2023 – 2027 (Expected)

B.Sc. in Software Industry and Multimedia (SIM)

Alexandria, Egypt

Languages & Activities

- **Languages:** Arabic (Native), English (Professional Working Proficiency)
- **Achievements:** Participant, Creativa × Orange AI Hackathon 2026 (shipped and deployed Kidney-RAG)
- **Volunteering:** Volunteer, Maaedat Farha